

The clustering lottery: initialisation variance in mixture-model clustering confounds spatial domain benchmarks

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Abstract

Motivation: Dozens of graph-based deep learning methods for spatial domain identification are compared on the same twelve human DLPFC Visium sections, and new methods are usually claimed to improve the adjusted Rand index (ARI) by a few hundredths. Almost all of them obtain domains by fitting a Gaussian mixture model (GMM, typically `mclust` EEE) to the learned embedding once. How much of the reported differences is due to the encoder, and how much to that final clustering step, is rarely measured.

Results: We re-implemented the recently published hierarchical graph VAE HiSTaR from its official code and ran it, a training-free graph-diffusion baseline (AnisoST), BANKSY and SpaGCN on all twelve DLPFC sections with three seeds and a held-out-donor protocol. (i) With the authors’ code and defaults, HiSTaR reached a median ARI of 0.509, not the reported 0.65. (ii) On a *fixed* embedding, refitting the same GMM from different random starts changed ARI by 0.21 on average and by up to 0.44 within one section; choosing the luckiest of 20 fits would raise the mean ARI by about 0.10, more than the gap between most published methods. (iii) When every method was clustered the same way (20 starts, best likelihood), HiSTaR (mean ARI 0.544), BANKSY (0.518) and a 3-second training-free baseline (0.551) were statistically indistinguishable; only SpaGCN remained clearly worse (0.442). We recommend that benchmarks report the clustering protocol, use multiple starts and seeds, and compare encoders under matched clustering.

Availability: Code, raw per-run results and figures: [TODO: repository URL / Zenodo DOI].

1 Introduction

Spatial transcriptomics assays such as 10x Visium measure genome-wide expression at thousands of spots while retaining their position in the tissue. Identifying *spatial domains*, i.e. regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian models [1] and neighbourhood-augmented features [2] to graph neural networks [3–5] and, recently, hierarchical graph variational autoencoders such as HiSTaR [6].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [7], and new methods typically report median ARI improvements of 0.02–0.10 over the previous state of the art. A large share of them share the same last step: the learned embedding is clustered with `mclust`’s EEE Gaussian mixture [8], fitted once. Independent benchmarks have found that method rankings depend strongly on the dataset [9, 10], that deep methods vary with the random seed [10], that replacing each method’s clustering algorithm changes results [11], and, most recently, that preprocessing and clustering choices often matter more than architectural novelty [12]. Simple, training-free pipelines have also been shown to be competitive [13, 14].

This paper is a focused, fully reproducible case study of that last point. Starting from an attempt to reproduce and improve HiSTaR, we ask three concrete questions: (1) Do HiSTaR’s published

DLPFC results reproduce with the official code? (2) On a fixed embedding, how much does ARI depend on the initialisation of the GMM alone? (3) Once every method uses the same, more robust clustering, do the differences between encoders remain?

2 Methods

2.1 Data and preprocessing

We used the twelve DLPFC Visium sections (three donors; seven annotated layers, five for donor 2) with the `layer_guess_reordered` annotation [7]. Every method received identical inputs: genes expressed in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour, on counts), expression was library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). The true number of domains was given to every method. Unannotated spots were excluded from evaluation.

2.2 Methods compared

- **HiSTaR** [6]: a faithful PyTorch port of the official code with the published defaults (cross-level weight 0.3, 200+200 epochs). Its output embedding was clustered as described below.
- **AnisoST** (this work, a reference baseline): edge-preserving graph diffusion [15] of the top 30 PCs on the spatial 6-nearest-neighbour graph,

$$w_{ij}^{(t)} = \exp\left(-\frac{\|h_i^{(t)} - h_j^{(t)}\|^2}{\kappa^{(t)}}\right), \quad H^{(t+1)} = (1 - \alpha)H^{(t)} + \alpha D^{(t)-1} W^{(t)} H^{(t)}, \quad (1)$$

with $\kappa^{(t)}$ the median squared edge difference, $\alpha = 0.5$ and $T = 10$ steps. It has no trainable parameters and is deterministic. Edge-preserving diffusion for spatial transcriptomics has been used before inside a deep model [16]; we use it only as a strong, cheap baseline. All settings were fixed on donor 1 only.

- **Linear diffusion**: the same with $w_{ij} = 1$ (ablation).
- **BANKSY** [2]: official `pybanksy` package, 18 neighbours, $m_{\max} = 1$, $\lambda \in \{0.2, 0.8\}$, 20 PCs of the BANKSY matrix.
- **SpaGCN** [3]: official package and tutorial pipeline without histology ($p = 0.5$, Louvain resolution search to the true number of domains, 200 epochs, with and without its hexagonal refinement). SpaGCN performs its own clustering.
- **HiCAST**, an extension of HiSTaR we designed earlier, and **PCA** + GMM (non-spatial) are reported for completeness.

2.3 Clustering protocols

All embeddings were clustered with a tied-covariance GMM, which is the same model as `mclust` EEE, as implemented in `scikit-learn` [17] (`reg_covar` = 10^{-5}). *Standard*: 3 random starts and a fixed seed, mimicking the usual single `mclust` call. *Robust*: 20 random starts; the fit with the highest log-likelihood is kept, which uses no labels. For the *lottery* experiment, one embedding per section and method (seed 0) was clustered 20 times with a single random start each (random states 0–19).

2.4 Evaluation and statistics

ARI [18] against the manual layers (NMI and FMS gave the same conclusions and are in the code repository). Each (method, section) pair was run with seeds 0, 1, 2; we report the mean over sections of the per-section mean, with 95% bootstrap confidence intervals over sections (5000 resamples). Methods were compared with two-sided Wilcoxon signed-rank tests on the twelve per-section means, with Holm correction [19] over the eight comparisons against AnisoST. Runtimes are single-thread CPU wall-clock times.

3 Results

3.1 HiSTaR’s published results did not reproduce

With the official code and published defaults, HiSTaR reached a median (over sections) ARI of 0.509 and a mean of 0.491 (95% CI 0.454–0.525), compared with a reported median of 0.65 [TODO: verify against the HiSTaR paper; also its reported 0.718 vs our 0.535 on 151673]. Re-running our port with the same seeds reproduced these values exactly. Fixing a bug that applies random masking at inference time did not change accuracy (0.487). Plausible reasons for the gap include per-section tuning of the cross-level loss weight (described in the HiSTaR supplement), unpublished preprocessing, R mclust vs. our equivalent GMM, and the choice of seed. The rest of the paper shows that the last two alone can account for differences of this size. [TODO: Contact the HiSTaR authors before submission.]

3.2 On a fixed embedding, the clustering step alone moves ARI by up to 0.44

Figure 1 shows, for every section, the ARI of 20 single-start GMM fits on one fixed embedding. For HiSTaR, the range of ARI within a section was 0.211 on average (0.061–0.439); on section 151669 the same embedding gave anything from 0.23 to 0.67. AnisoST embeddings, which are deterministic, were just as sensitive (mean range 0.227, maximum 0.422). The mean ARI of a single random fit was 0.489 (HiSTaR) and 0.500 (AnisoST), whereas the best of the 20 fits by ARI averaged 0.596 and 0.601: selecting the clustering run by its agreement with the labels, explicitly or by trying a few seeds, would inflate the reported mean ARI by about 0.10.

Keeping the fit with the best log-likelihood, which uses no labels, recovered part of this: 0.546 (HiSTaR) and 0.560 (AnisoST). Likelihood and ARI were only weakly related, especially for HiSTaR (median Spearman ρ across sections 0.22, positive in 7 of 12 sections; AnisoST 0.48, positive in 11 of 12), so more restarts reduce but do not remove the dependence on luck.

3.3 Under matched clustering, the encoders are indistinguishable

Table 1 and Figure 2 compare the methods under both clustering protocols. With standard clustering, AnisoST appeared better than HiSTaR by 0.060 mean ARI (9/12 sections; $p = 0.012$, Holm-adjusted $p = 0.061$). When HiSTaR’s embedding was clustered with the same robust protocol, its mean ARI rose from 0.491 to 0.544 (8/12 sections; $p = 0.077$) and the difference to AnisoST shrank to 0.006 ($p = 0.47$). BANKSY with robust clustering (0.518 for both λ) and plain linear diffusion (0.558) were also statistically indistinguishable from AnisoST (all Holm-adjusted $p = 1$). SpaGCN (0.442 refined), HiCAST (0.482) and the non-spatial baseline (0.409) remained significantly worse (Holm-adjusted $p \leq 0.03$).

Better clustering helped every embedding but by different amounts (+0.053 HiSTaR, +0.024 AnisoST, +0.024 BANKSY $\lambda = 0.8$, +0.004 BANKSY $\lambda = 0.2$). The ranking of methods under

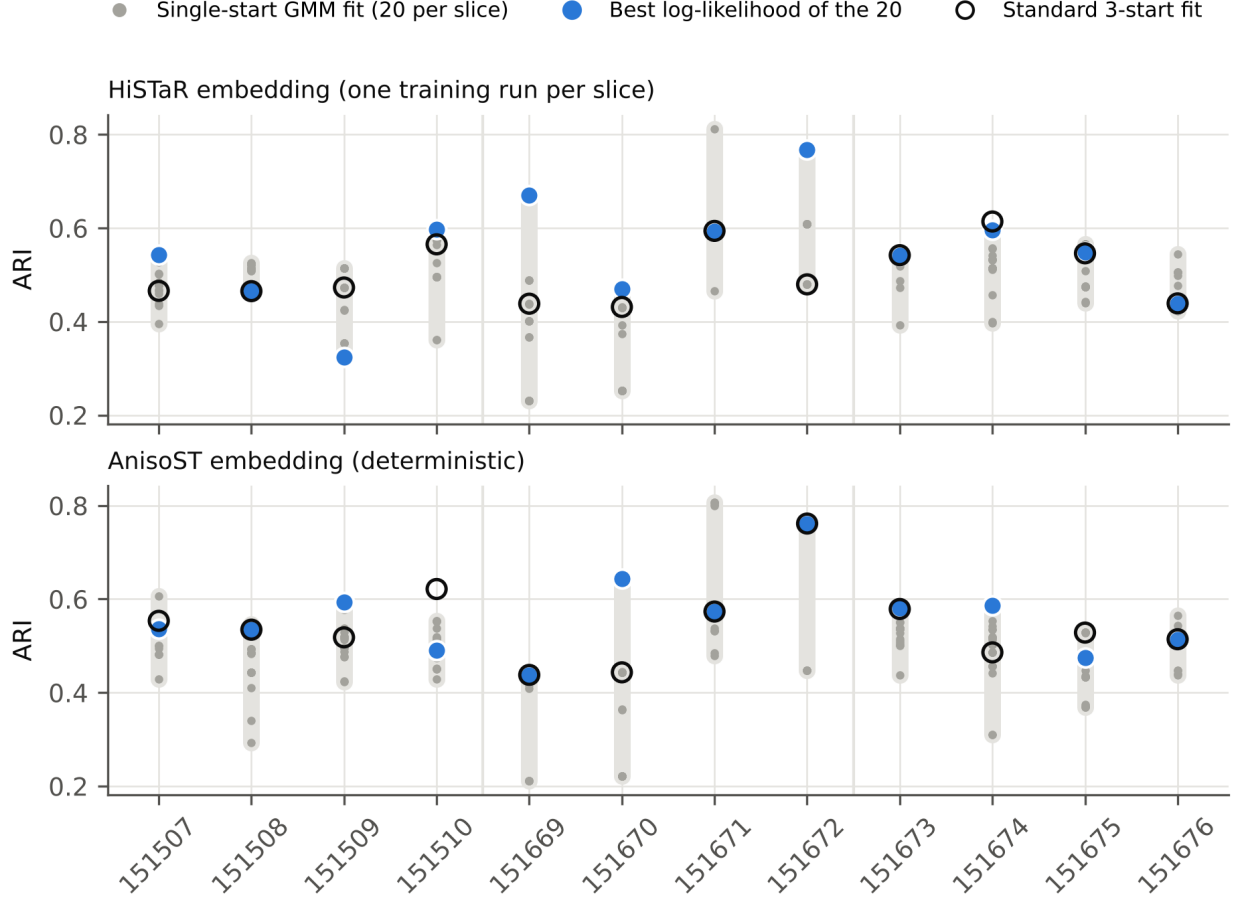


Figure 1: The clustering lottery. For each DLPFC section, one embedding was clustered 20 times with a single-start tied-covariance GMM (grey dots; grey bar = range). Blue: the fit with the highest log-likelihood (label-free); open circle: the standard 3-start fit.

standard clustering therefore partly reflects how well each embedding suits a single short GMM fit, not only its quality.

The same pattern held on the two held-out donors, where HiSTaR with robust clustering was slightly ahead (0.569 vs. 0.554; Table 1). The main remaining differences were cost and stability: AnisoST took 3.4 s per section against 65–112 s for HiSTaR, and its mean seed-to-seed standard deviation was 0.020 against 0.067 for HiSTaR with robust clustering, whose remaining variance comes from training.

3.4 A training-free baseline

Graph smoothing was the ingredient that mattered: diffusion improved the non-spatial baseline by 0.142 ARI (12/12 sections). Edge preservation did not raise ARI over linear diffusion, but it improved accuracy on boundary spots (spots with a neighbour in another layer, 14–30% of annotated spots) by 0.025 (9/12 sections, $p = 0.034$, uncorrected) without changing interior accuracy (Supplementary Fig. S2). All 23 one-at-a-time hyper-parameter settings we tried gave a mean ARI between 0.531 and 0.570 (Supplementary Fig. S1). Figure 3 shows an example section.

Table 1: Twelve DLPFC sections $\times$ 3 seeds. Mean ARI over sections with 95% bootstrap CI; held-out = donors 2 and 3 (AnisoST settings were fixed on donor 1). Δ = AnisoST minus method; p_{Holm} from Wilcoxon signed-rank tests over sections. Seed SD = mean over sections of the SD across seeds.

Method	Mean ARI [95% CI]	Held-out	Seed SD	Time (s)	Δ	p_{Holm}
<i>Robust clustering (20 starts, best likelihood)</i>						
AnisoST (baseline)	0.551 [0.509, 0.599]	0.554	0.020	3.4	—	—
Linear diffusion	0.558 [0.494, 0.633]	0.577	0.007	3.0	−0.007	1.00
HiSTaR	0.544 [0.492, 0.608]	0.569	0.067	65	+0.006	1.00
BANKSY $\lambda = 0.8$	0.518 [0.461, 0.575]	0.531	0.046	5.4	+0.033	1.00
BANKSY $\lambda = 0.2$	0.518 [0.454, 0.580]	0.522	0.040	5.4	+0.032	1.00
PCA (non-spatial)	0.409 [0.351, 0.470]	0.428	0.026	3.5	+0.142	0.004
<i>Standard clustering (3 starts) or method's own</i>						
HiSTaR	0.491 [0.454, 0.525]	0.497	0.047	112	+0.060	0.061
HiCAST	0.482 [0.423, 0.534]	0.470	0.043	245	+0.069	0.029
SpaGCN (refined)	0.442 [0.390, 0.489]	0.443	0.040	42	+0.109	0.007

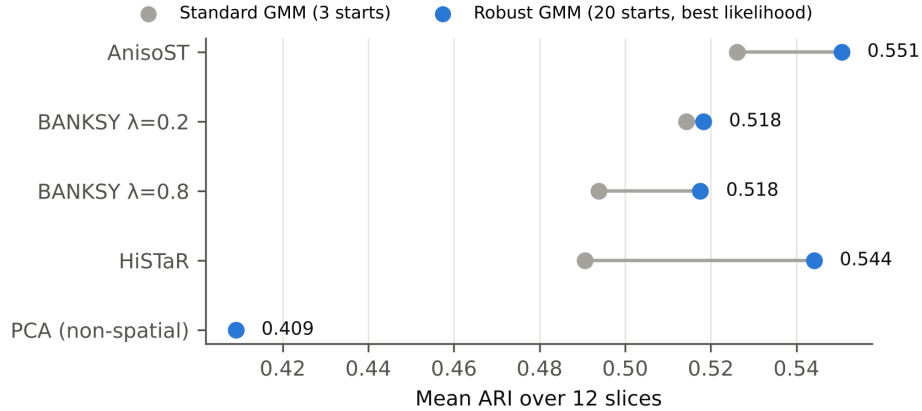


Figure 2: Mean ARI over the twelve sections with the standard 3-start GMM (grey) and the 20-start, best-likelihood GMM (blue) applied to the same embeddings.

4 Discussion

On the most widely used benchmark for spatial domain identification, the clustering step alone produces differences in ARI as large as those attributed to new architectures. A fixed embedding can score anywhere in a 0.2–0.4 ARI range depending on the GMM’s random start, and choosing the best of a handful of runs could inflate a reported result by about 0.10. Once clustering was matched, a hierarchical graph VAE, BANKSY and a training-free diffusion baseline were statistically indistinguishable. This agrees with large recent benchmarks [10–12] and adds a direct, quantitative measurement of initialisation variance on fixed embeddings.

Recommendations. (1) Report the clustering protocol (algorithm, number of starts, seed, and how it was chosen). (2) Use multiple starts and select by likelihood, never by ARI. (3) Report results over several training seeds *and* clustering seeds, with per-section values. (4) Compare new encoders against strong training-free baselines under the same clustering. (5) Report runtime and

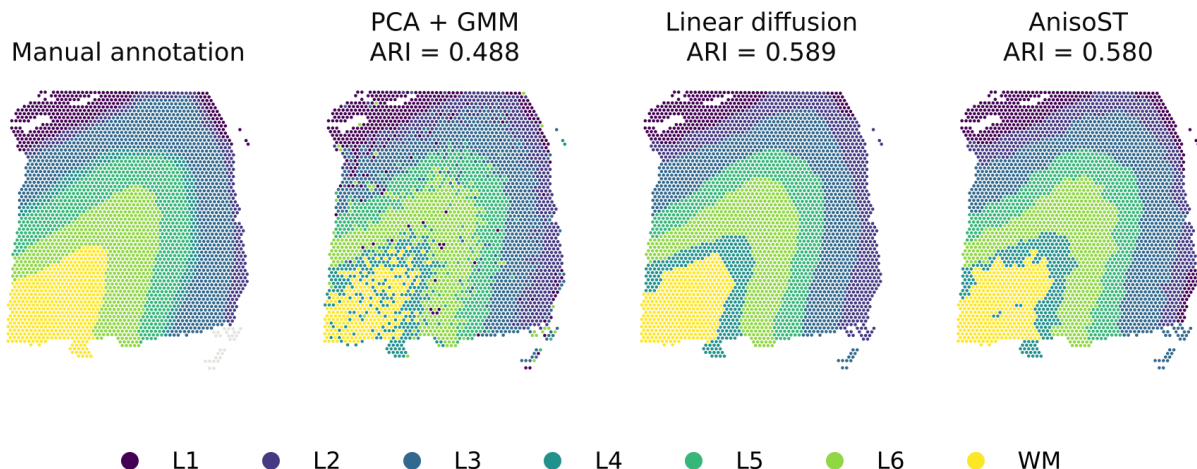


Figure 3: Section 151673: manual annotation and domains from non-spatial PCA + GMM, linear diffusion and AnisoST (robust clustering).

seed variance alongside accuracy.

Limitations. This study uses one dataset (DLPFC), six methods and one embedding per method for the lottery experiment. mclust initialises EM with model-based hierarchical agglomeration rather than random starts, so the exact variance of a single mclust call may differ from our random-start GMM [TODO: repeat the lottery with R mclust, e.g. on subsampled data or perturbed embeddings]. We could not reproduce HiSTaR’s per-section tuning, and we did not evaluate GraphST, STAGATE, SEDR or BayesSpace. The true number of domains was given to all methods. [TODO: Add at least one dataset with ground truth (e.g. 10x breast cancer, MERFISH, STARmap).]

Data and code availability

DLPFC count matrices are from the spatialLIBD project; layer annotations from the SEDR analyses repository. All code (preprocessing, HiSTaR port, AnisoST, baselines, figures) and the raw per-run results are available at [TODO: URL].

Use of AI tools

[TODO: Describe per journal policy: code, experiments and the first draft of this manuscript were produced with the assistance of an AI coding assistant (Claude, Anthropic); all results were checked by the authors, who take full responsibility for the content.]

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[TODO: Funding.] The authors declare no conflict of interest.

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Supplementary material

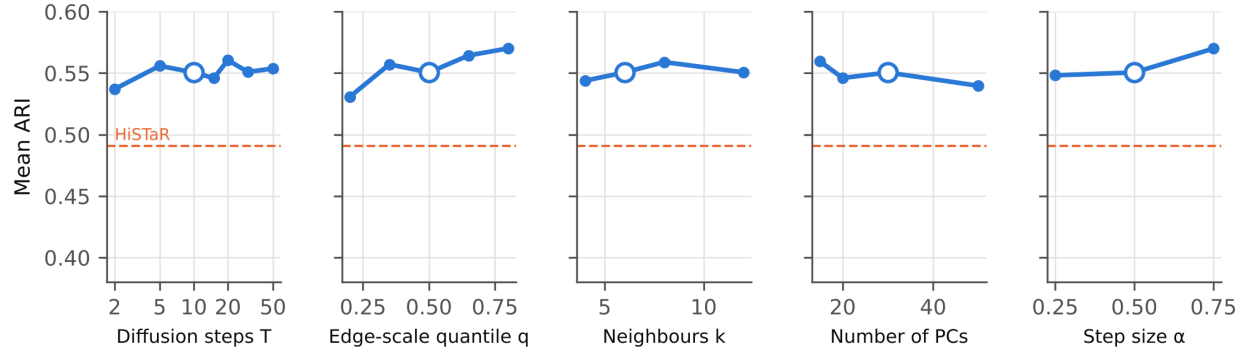


Figure S1: AnisoST mean ARI (12 sections $\times$ 3 seeds) when one hyper-parameter is changed at a time; open circle = default; dashed = HiSTaR, standard clustering.

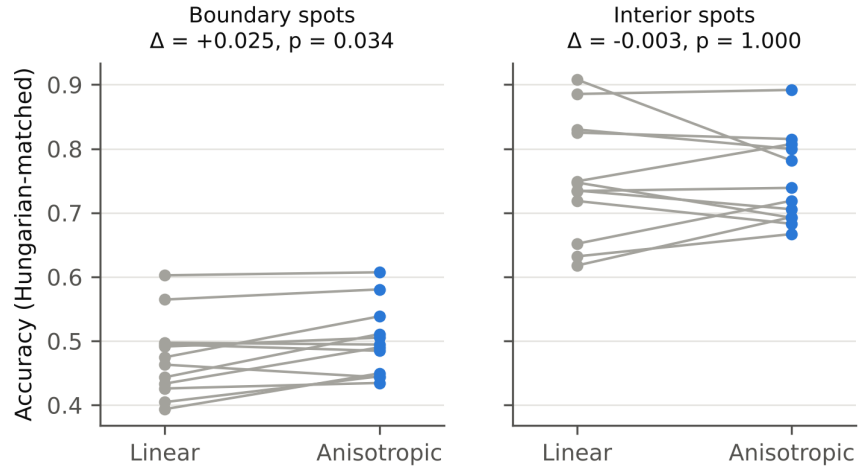


Figure S2: Accuracy (Hungarian-matched) on boundary and interior spots, linear vs. edge-preserving diffusion; one line per section (mean of 3 seeds).

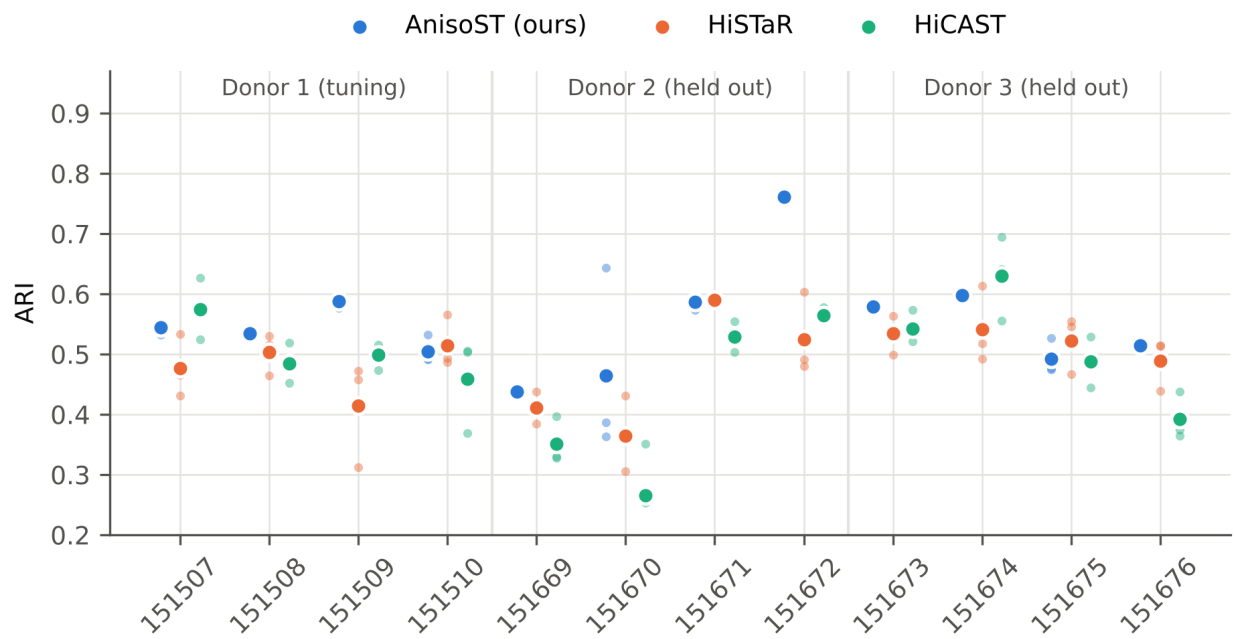


Figure S3: Per-section ARI of AnisoST (robust clustering), HiSTaR and HiCAST (standard clustering); small dots are individual seeds.